

# BAUST - Department of Electrical & Electronic Engineering

EEE 1201: Electrical Circuit Analysis | Class Test 02 | Level 1 / Term 2

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**Time: 45 Minutes | Full Marks: 20**

## **Question 1: Thevenin and Norton Theorems [10 Marks]**

Find the Thevenin equivalent circuit across terminals A-B for a circuit with 12V DC source, 4 ohm resistor in series, parallel 6 ohm resistor, and dependent voltage source  $2V_x$ .

## **Question 2: Sinusoidal AC Steady-State Analysis [10 Marks]**

An RLC circuit with  $R = 10\ \Omega$ ,  $L = 50\ \text{mH}$ ,  $C = 100\ \mu\text{F}$  is driven by  $v(t) = 100 \cos(1000t + 30^\circ)$ .

- Verified Academic Document, Peer's Charity Academic Repository, Contributed by Nafis Iqbal
- Calculate total circuit impedance in phasor notation (magnitude and phase angle).
  - Determine average active power delivered to the circuit and power factor.